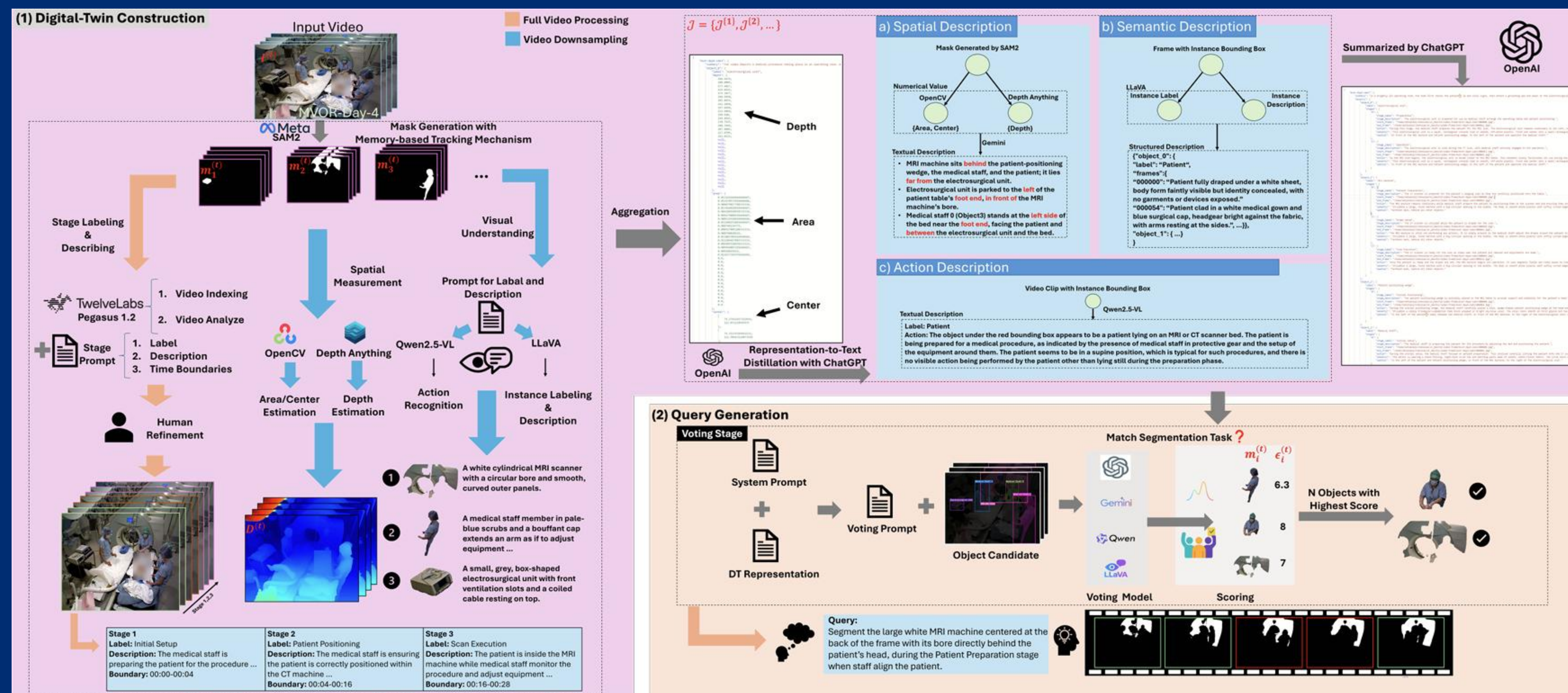
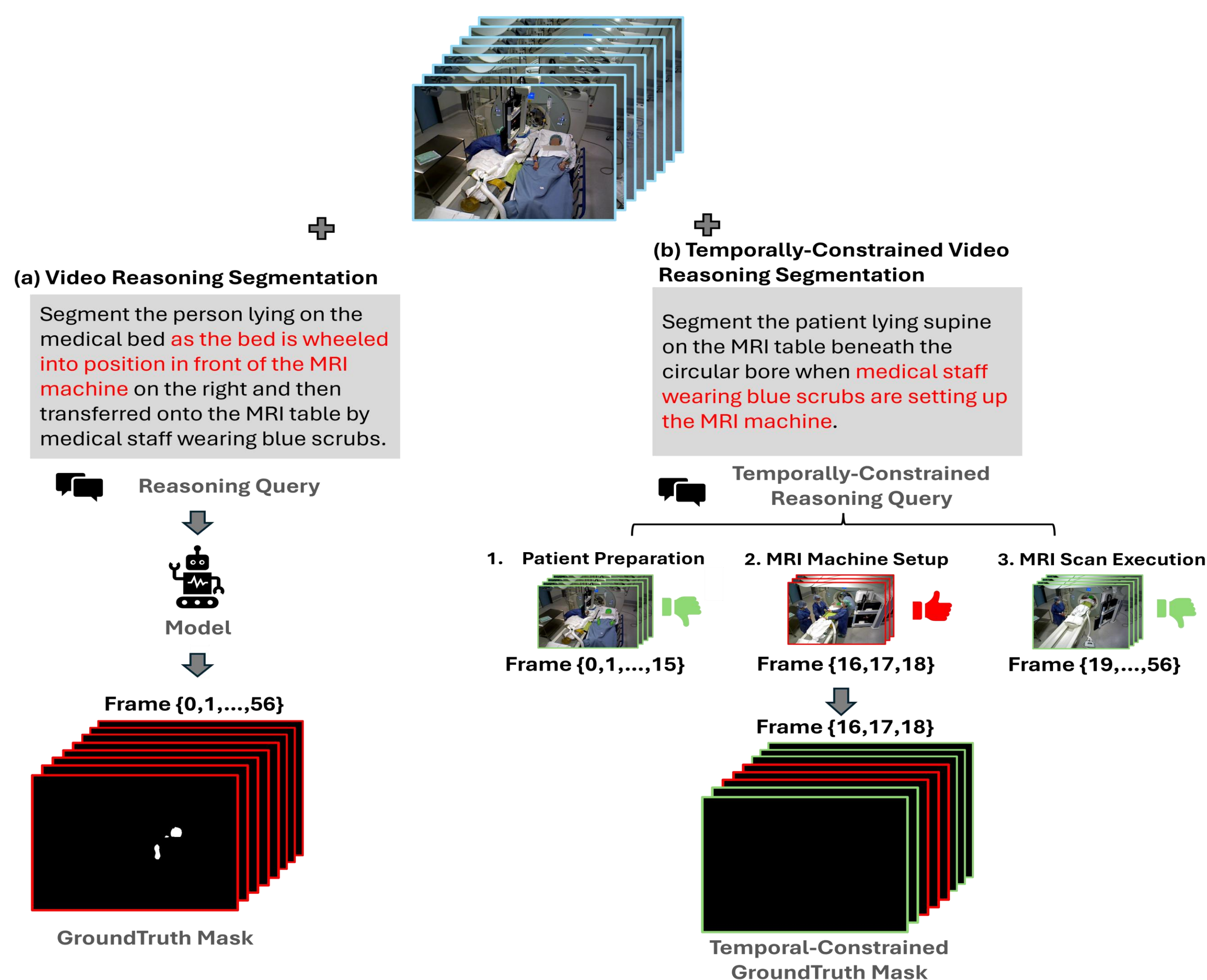


Temporally-Constrained Video Reasoning Segmentation and Automated Benchmark Construction

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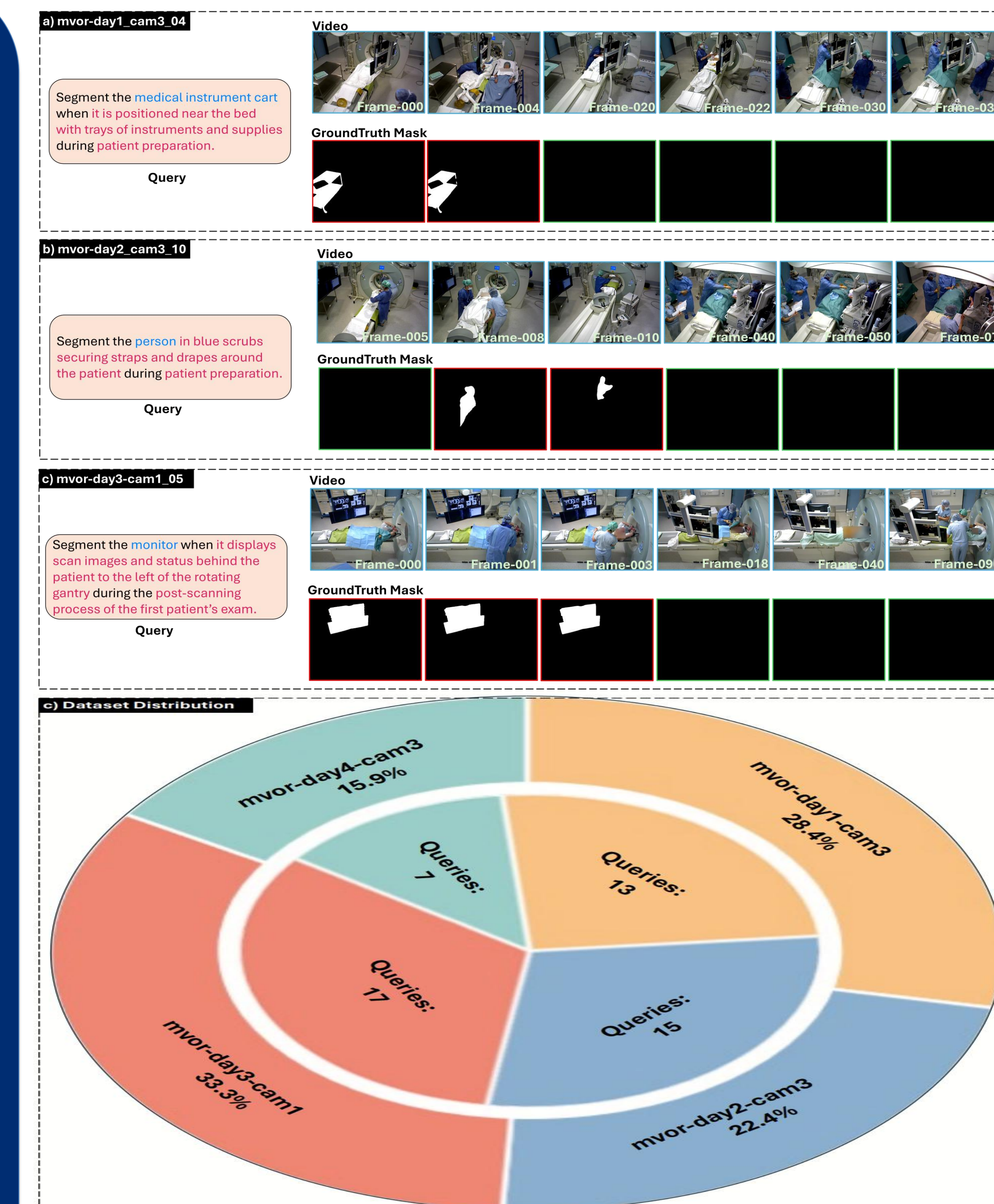
Existing video Reasoning Segmentation methods have limitations in temporal relevance, assuming target objects remain relevant throughout entire videos.



The automated benchmark construction method uses a digital twin representation that preserves semantic, spatial, and temporal relationships to generate temporally constrained queries and masks.

Key conclusions:

- **Task Reformulation:** Temporally-constrained RS removes the assumption that objects stay relevant throughout the whole video.
- **Scalability:** The automated benchmark construction method allows creation of TCVideoRSBenchmark without heavy manual annotations.



- **Dynamic Nature of Real Scenarios:** Captures temporal dynamism.
- **Query Diversity:** Covers semantic, spatial, and temporally constrained queries to evaluate multiple reasoning dimensions.
- **Quality Assurance:** Ensured through digital twin representation.

Our method introduces temporal boundaries into video RS tasks with TCVideoRSBenchmark via automated generation to improve when queries are contextually relevant.